

Wu WenXuan

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EDUCATION

2024.9 - 2028.6 **Fudan University** Shanghai

B.S. in Mathematics and Applied Mathematics (Mathematics Talent Program), expected June 2028

- **GPA / Honors:** 3.95 / 4.0 (Rank: 1 / 99); Fudan University Scholarship.
- **Selected Mathematics Coursework:** Mathematical Analysis (A), Advanced Linear Algebra (A), Classical Mathematical Thought (A), Ordinary Differential Equations (A+).

COMPETITION AWARDS

- American Invitational Mathematics Examination (AIME): Distinction (2024); near-perfect score 14 / 15 (2023) in the invitation-only contest.
- AMC 12: Certificate of Distinction (2022); Honor Prize (2023), demonstrating sustained high-level mathematical problem-solving.
- China High School Mathematics League (Shanghai Region): Third Prize (2023); WMTC Individual Excellence Award (2022).
- Harvard China Thinks Big (CTB): National First Prize, Innovation & Research Track (2023).

EXPERIENCE

2026.1 - Present **Quant Info Tech (量锐科技)** | *Strategy Intern, Rates & Fixed Income Group* Shanghai

- **Bond Pair Trading:** Researched yield-spread mean-reversion and relative-value logic for bond pair strategies; validated hypotheses with Rates AI and Bond Seek.

2025.6 - 2025.8 **Shanghai Silver Leaf Investment (银叶投资)** | *Quantitative Research Intern, Strategy Department* Shanghai

- **Style Rotation:** Built small-cap vs. large-cap timing signals from crowding indicators; applied TDA / Persistent Homology to detect inflections, with backtest SR > 1.0.

SELECTED QUANTITATIVE PROJECTS

2026.5 **Transformer-based Price Prediction for High-Volatility Assets** | *Lead Researcher*

- **Model Design:** Built a state-routed Dual-Expert Transformer for minute-level forecasting across normal and extreme price-formation regimes.
- **Results:** Applied Scale Normalization, Power-Log targets and rolling TTA; achieved Mean IC 0.0871, with Crypto extreme-regime Mean IC 0.1018.

2026.3 - 2026.4 **High-Frequency Stock Return Prediction Model (Guotai Haitong Competition)** | *Team Leader*

- **Signal Fusion:** Built an E-stream / Board-stream Ridge framework combining micro-order-book and sector spillover signals for low-SNR tick data.
- **Validation:** Designed dynamic leader-stock selection and leakage-safe Walk-Forward filtering; delivered +0.0055 absolute Mean IC over baselines.

2023.7 - 2024.7 **Domestic Macro Asset Dynamic Allocation Model** | *Independent Lead*

- **Optimization:** Derived analytical gradients and Jacobians for rolling Markowitz optimization, cutting 17-year backtest runtime from nearly 1 hour to 90 seconds (~40x).
- **Risk Model:** Introduced EWMA covariance forecasting; achieved Return/Risk ratio of 1.0 across 2006-2023 vs. 0.79 for the fixed-income benchmark.

SKILLS & INTERESTS

- Mathematical Interests: probability, optimization, market microstructure, machine learning on noisy data, and puzzle-style problem solving.
- Programming & Tools: Python, LaTeX, SPSS; quantitative research workflows for noisy market data.
- Languages: English (IELTS 7.5), Mandarin (Native).
- Activities: Fudan University Math Student Union, Minister of Academic and Innovation Department; curated MathHall lectures.